Smart Sorting: A Machine Learning Approach to Identifying Rotten Fruits and Vegetables

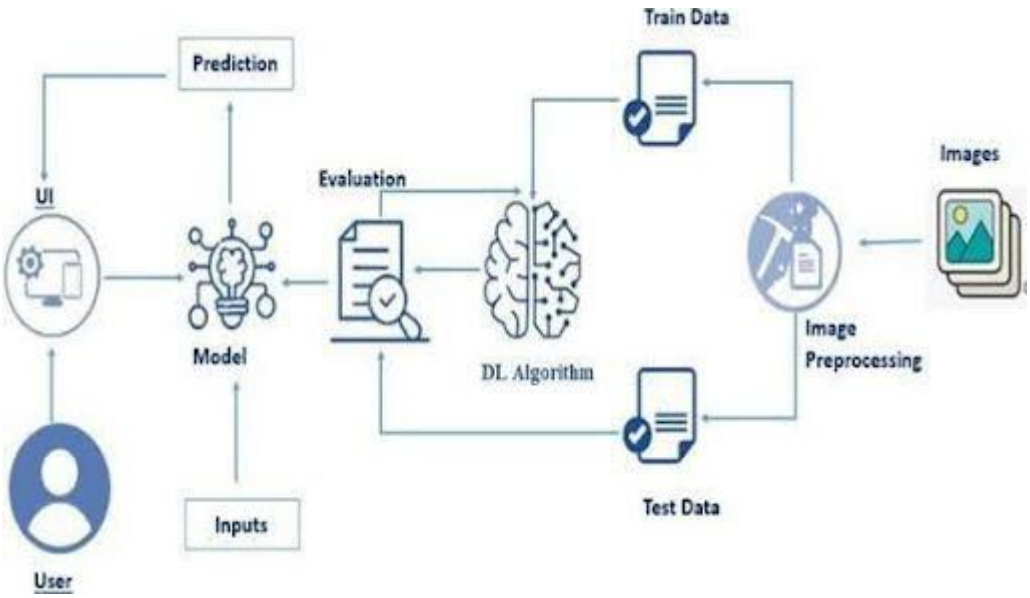
Project Description

Food spoilage is a major concern in households, markets, and supply chains, leading to economic loss, health risks, and unnecessary food waste. Identifying whether fruits and vegetables are fresh or rotten is often done manually, which can be inconsistent and time-consuming. With increasing demand for automated quality inspection, machine learning–based systems have become essential for improving accuracy and reliability in freshness detection.

Fruit freshness classification is challenging because spoilage indicators depend on subtle visual factors such as color variation, texture changes, and surface irregularities. Analyzing image data allows intelligent systems to learn patterns that distinguish fresh produce from spoiled items. Early and accurate detection helps consumers, vendors, and quality inspectors make informed decisions, improving food safety and reducing waste.

In this project, a deep learning image classification model based on transfer learning is trained using labeled fruit and vegetable images. The model learns to extract meaningful visual features and classify items as Fresh or Rotten. After performance evaluation, the best-performing model is integrated into a Flask-based web application, where users can upload an image and instantly receive a freshness prediction.

Technical Architecture:



Pre requisites:

- To complete this project, you must require the following software, concepts, and packages
    - Anaconda Navigator:
        - Refer to the link below to download Anaconda Navigator
    - Python packages:
    - Open anaconda prompt as administrator
    - Type “pip install numpy” and click enter.
    - Type “pip install pandas” and click enter.
    - Type “pip install scikit-learn” and click enter.
    - Type ”pip install matplotlib” and click enter.
    - Type ”pip install scipy” and click enter.
    - Type ”pip install seaborn” and click enter.
    - Type ”pip install tenserflow” and click enter.
    - Type “pip install Flask” and click enter.

Prior Knowledge:

- You must have prior knowledge of the following topics to complete this project.
- DL Concepts
- Neural Networks:: https://www.analyticsvidhya.com/blog/2020/02/cnn-vs-rnn-vs-mlp-analyzing-3-types-of-neural-networks-in-deep-learning/
- Deep Learning Frameworks:: https://www.knowledgehut.com/blog/data-science/pytorch-vs-tensorflow
- Transfer Learning: https://towardsdatascience.com/a-demonstration-of-transfer-learning-of-vgg-convolutional-neural-network-pre-trained-model-with-c9f5b8b1ab0a
- VGG16: https://www.geeksforgeeks.org/vgg-16-cnn-model/
- Convolutional Neural Networks (CNNs): https://www.analyticsvidhya.com/blog/2021/05/convolutional-neural-networks-cnn/ s://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-

learning
- Overfitting and Regularization: https://www.analyticsvidhya.com/blog/2021/07/prevent-overfitting-using-regularization-techniques/
- Optimizers: https://www.analyticsvidhya.com/blog/2021/10/a-comprehensive-guide-on-deep-learning-optimizers/
- Flask Basics: https://www.youtube.com/watch?v=lj4I_CvBnt0
-

Project Objectives:

By the end of this project you will:

- Know fundamental concepts and techniques used for machine learning.
- Gain a broad understanding about data.
- Have knowledge on pre-processing the data/transformation techniques on outlier and some visualization concepts.

Project Flow:

- User interacts with the UI to enter the input.
- Entered input is analyzed by the model which is integrated.
- Once model analyses the input the prediction is showcased on the UI

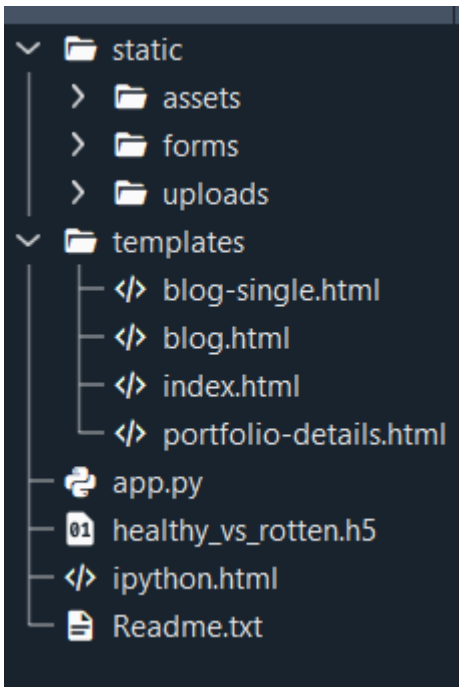
To accomplish this, we have to complete all the activities listed below,

- Data collection
  - Collect the dataset or create the dataset
- Visualizing and analyzing data
  - Univariate analysis
  - Bivariate analysis
  - Multivariate analysis
  - Descriptive analysis
- Data pre-processing
  - Checking for null values
  - Handling outlier
  - Handling categorical data
  - Splitting dataset into Dependent And Independent Variables
  - Splitting data into train and test
  - Feature Scaling
- Model building
  - Import the model building libraries
  - Initializing the model
  - Training and testing the model
  - Evaluating performance of model

    - Save the model
- Application Building
    - Create an HTML file
    - Build python code

Project Structure:

Create the Project folder which contains files as shown below



- We are building a flask application which needs HTML pages stored in the templates folder and a python script app.py for scripting.
- floods.pkl is our saved model. Further we will use this model for flask integration.
- Training folder contains model training files and training_ibm folder contains IBM deployment files.

Milestone 1: Data Collection

ML depends heavily on data, It is most crucial aspect that makes algorithm training possible. So this section allows you to download the required dataset.

Activity 1: Download the dataset

There are many popular open sources for collecting the data. Eg: kaggle.com, UCI repository, etc.

In this project we have used Floods_Data.csv data. This data is downloaded from kaggle.com. Please refer to the link given below to download the dataset.

Link: https://www.kaggle.com/datasets/arbethi/rainfall-dataset

Milestone 2: Visualizing and analysing the data

As the dataset is downloaded. Let us read and understand the data properly with the help of some visualization techniques and some analysing techniques.

Note: There is n number of techniques for understanding the data. But here we have used some of it. In an additional way, you can use multiple techniques.

Activity 1: Importing the libraries

Import the necessary libraries as shown in the image. (optional) Here we have used visualization style as fivethirtyeight.

```
import os
import shutil
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from sklearn.model_selection import train_test_split
import shutil
from sklearn.model_selection import train_test_split
from tensorflow.keras.applications.vgg16 import VGG16
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.models import Model
from keras.optimizers import Adam
from tensorflow.keras.applications.resnet50 import ResNet50
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.models import Model
from keras.preprocessing import image
from keras.applications.vgg16 import preprocess_input
from tensorflow.keras.preprocessing.image import load_img, img_to_array
```

Activity 2: Read the Dataset

Our dataset format might be in .csv, excel files, .txt, .json, etc. We can read the dataset with the help of pandas.

In pandas we have a function called read_csv() to read the dataset. As a parameter we have to give the directory of csv file.

```
!mkdir ~/.kaggle

!cp kaggle.json ~/.kaggle

!kaggle datasets download -d muhammad0subhan/fruit-and-vegetable-disease-healthy-vs-rotten
```

```
Warning: Your Kaggle API key is readable by other users on this system! To fix this, you can run 'chmod 600 /root/.kaggle/kaggle.json'
Dataset URL: https://www.kaggle.com/datasets/muhammad0subhan/fruit-and-vegetable-disease-healthy-vs-rotten
License(s): CC0-1.0
Downloading fruit-and-vegetable-disease-healthy-vs-rotten.zip to /content
100% 4.76G/4.77G [00:41<00:00, 136MB/s]
100% 4.77G/4.77G [00:41<00:00, 124MB/s]
```

```
!unzip  /content/fruit-and-vegetable-disease-healthy-vs-rotten.zip
```

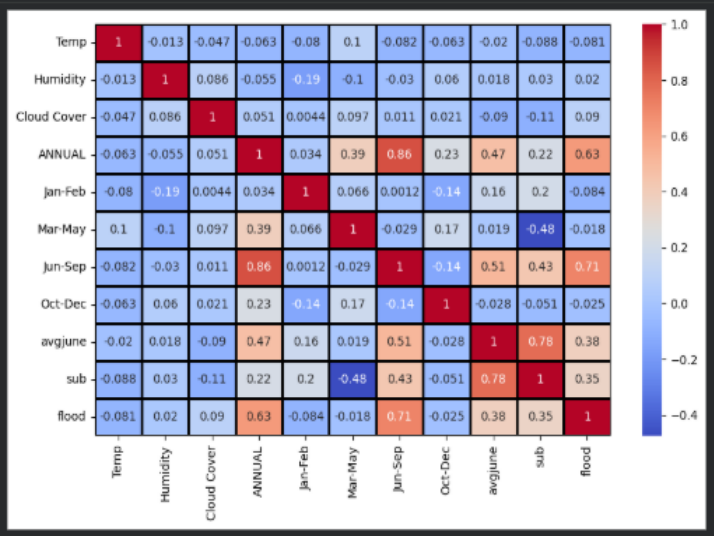
Activity 3: Univariate analysis

In simple words, univariate analysis is understanding the data with single feature. Here we have displayed two different graphs such as distplot and countplot.

- Seaborn package provides a wonderful function distplot. With the help of distplot, we can find the distribution of the feature. To make multiple graphs in a single plot, we use subplot.

From the above graph, we can infer that the column temp follows some kind of normal distribution, it means the data is normal is almost normal

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm",
            linewidths=1, linecolor='k')
plt.show()
```



A heat map is a data visualization technique that shows magnitude of a phenomenon as colour in two dimensions. The variation in colour may be by hue or intensity, giving obvious visual cues to the reader about how the phenomenon is clustered or varies over space.

- From the graph, we can identify less correlated values and can drop those variables.

Activity 6: Descriptive analysis

Descriptive analysis is to study the basic features of data with the statistical process. Here pandas has a worthy function called describe. With this describe function we can understand the unique, top and frequent values of categorical features. And we can find mean, std, min, max and percentile values of continuous features.

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 115 entries, 0 to 114
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Temp         115 non-null    int64
 1   Humidity     115 non-null    int64
 2   Cloud Cover  115 non-null    int64
 3   ANNUAL       115 non-null    float64
 4   Jan-Feb      115 non-null    float64
 5   Mar-May      115 non-null    float64
 6   Jun-Sep      115 non-null    float64
 7   Oct-Dec      115 non-null    float64
 8   avgjune      115 non-null    float64
 9   sub          115 non-null    float64
 10  flood        115 non-null    int64
dtypes: float64(7), int64(4)
memory usage: 10.0 KB
```

	Temp	Humidity	Cloud Cover	ANNUAL	Jan-Feb	Mar-May	Jun-Sep	Oct-Dec	avgjune	sub	flood
count	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000	115.000000
mean	29.600000	73.852174	36.286957	2925.487826	27.739130	377.253913	2022.840870	497.636522	218.100870	439.801739	0.139130
std	1.122341	2.947623	4.330158	422.112193	22.361032	151.091850	386.254397	129.880643	62.547597	210.438813	0.347597
min	28.000000	70.000000	30.000000	2068.800000	0.300000	89.900000	1104.300000	166.600000	65.600000	34.200000	0.000000
25%	29.000000	71.000000	32.500000	2627.900000	10.250000	276.750000	1768.850000	407.450000	179.666667	295.000000	0.000000
50%	30.000000	74.000000	36.000000	2937.500000	20.500000	342.000000	1948.700000	501.500000	211.033333	430.600000	0.000000
75%	31.000000	76.000000	40.000000	3164.100000	41.600000	442.300000	2242.900000	584.550000	263.833333	577.650000	0.000000
max	31.000000	79.000000	44.000000	4257.800000	98.100000	915.200000	3451.300000	823.300000	366.066667	982.700000	1.000000

```
# This is formatted as code
```

As you can see in our dataset there is no textual data, all the set of data is in float and integer type.

Describe () functions are used to compute values like count, mean, standard deviation give a summary type of data.

Milestone 3: Data Pre-processing

As we have understood how the data is lets pre-process the collected data.

The download data set is not suitable for training the machine learning model as it might have so much of randomness so we need to clean the dataset properly in order to fetch good results. This activity includes the following steps.

- Handling missing values
- Handling categorical data
- Handling outliers
- Splitting dataset into Dependent And Independent Variables
- Splitting data into train and test
- Feature Scaling

Note: These are the general steps of pre-processing the data before using it for machine learning. Depending on the condition of your dataset, you may or may not have to go through all these steps.

Activity 1: Checking for null values

- For checking the null values, df.isnull() function is used. To sum those null values we use .sum() function to it. From the below image we found that there are no null values present in our dataset. So we can skip handling of missing values step.

```
#checking null values
dataset.isnull().any()
```

```
Temp           False
Humidity       False
Cloud Cover    False
ANNUAL         False
Jan-Feb        False
Mar-May        False
Jun-Sep        False
Oct-Dec        False
avgjune        False
sub            False
flood          False
dtype: bool
```

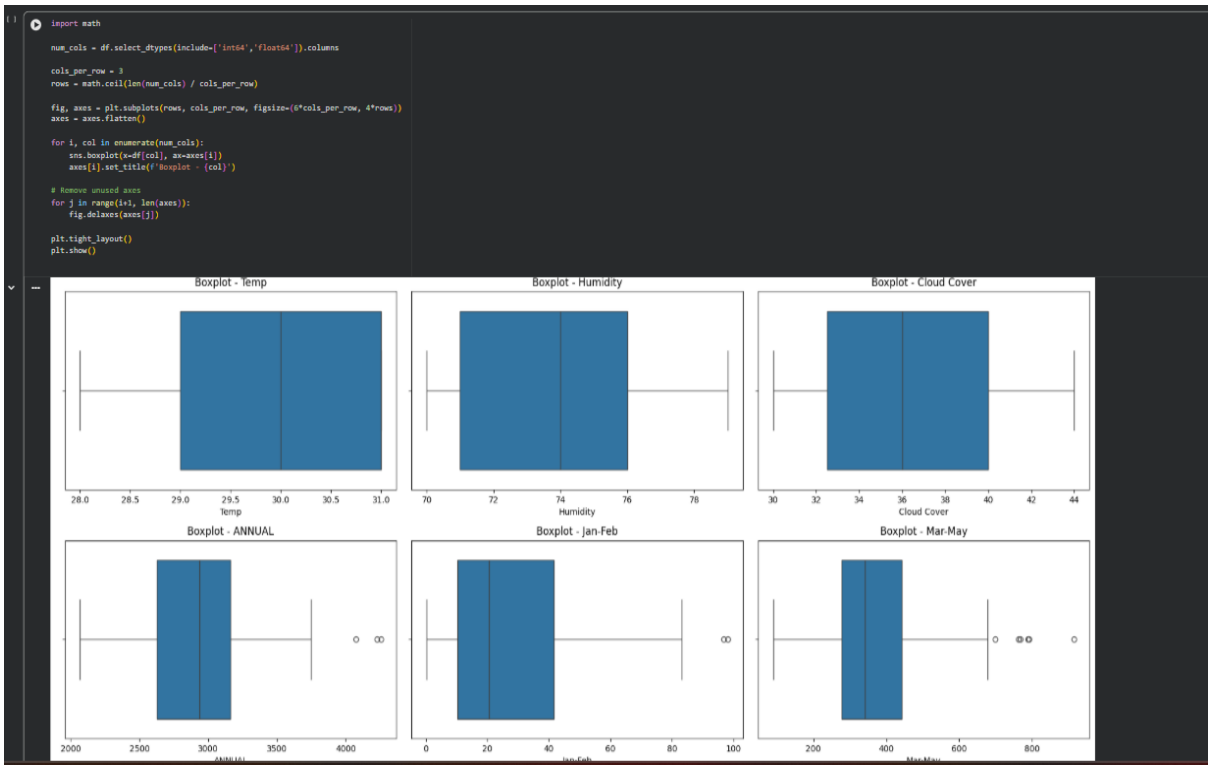
From the above code of analysis, we can infer that columns such as newbalanceOrg, oldbalnceOrig,isFraud are having the missing values, we need to treat them in a required way.

Activity 2: Handling Outliers

With the help of boxplot, outliers are visualized (refer activity 3 univariate analysis). And here we are going to find upper bound and lower bound of Na_to_K feature with some mathematical formula.

- To find the upper bound we have to multiply IQR (Interquartile range) with 1.5 and add it with 3rd quantile. To find lower bound instead of adding, subtract it with 1st quantile. Take image attached below as your reference.
- If outliers are removed, we lose more data. It will impact model performance.

- Here removing outliers is impossible. So, the capping technique is used on outliers.
- Capping: Replacing the outliers with upper bound values.



Note: In our Dataset all the values are in the same range, so outliers replacing is not necessary.

Activity 3: Handeling Categocal Values

As we can see our dataset has categorical data we must convert the categorical data to integer encoding or binary encoding.

To convert the categorical features into numerical features we use encoding techniques. There are several techniques but in our project, we are using feature mapping and label encoding.

Note: In our dataset, there is no categorical data type, so we can skip this step.

Activity 4: Splitting Dataset Into Dependent And Independent Variables

In machine learning, the concept of the dependent variable (y) and independent variables(x) is important to understand. Here, the Dependent variable is nothing but output in the dataset and the independent variable is all inputs in the dataset. We can denote with any symbol (alphabets). In our dataset, we can say that class is the dependent variable and all other columns are independent. But in order to select the independent columns, we will be selecting only those columns which are highly correlated and some value to our dependent column.

· With this in mind, we need to split our dataset into the matrix of independent variables and the vector or dependent variable. Mathematically, Vector is defined as a matrix that has just one column.

· **Let's create out independent and dependent variables:**

```
vgg16.save('healthy_vs_rotten')
```

Activity 5: Splitting data into train and test

Now let's split the Dataset into train and test sets

Changes: first split the dataset into x and y and then split the data set

```
trainpath = "/content/output_dataset/train"
testpath= "/content/output_dataset/test"

train_datagen = ImageDataGenerator(rescale = 1./255,zoom_range= 0.2,shear_range= 0.2)
test_datagen = ImageDataGenerator(rescale = 1./255)

train = train_datagen.flow_from_directory(trainpath,target_size =(224,224),batch_size = 20)
test = test_datagen.flow_from_directory(testpath,target_size =(224,224),batch_size = 20) ,#5 ,15 , 32, 50

Found 3358 images belonging to 28 classes.
Found 1120 images belonging to 28 classes.
```

Here x and y variables are created. On x variable, df is passed with dropping the target variable. And on y target variable is passed. For splitting training and testing data we are using train_test_split() function from sklearn. As parameters, we are passing x, y, test_size, random_state.

Activity 6: Feature Scaling

- Standard Scaler: Sklearn its main scaler, the StandardScaler, uses a strict definition of standardization to standardize data. It purely centers the data by using the following formula, where u is the mean and s is the standard deviation.

FEATURE SCALING

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Saving the standard Scaler

```
import joblib

joblib.dump(scaler, "transform.save")
```

```
['transform.save']
```

Milestone 4: Model Building

Now our data is cleaned and it's time to build the model. We can train our data on different algorithms. for this project we are applying four classification algorithms. The best model is saved based on its performance.

Activity 1: Decision tree model

A function named decisionTree is created and train and test data are passed as the parameters. Inside the function, DecisionTreeClassifier algorithm is initialized and training data is passed to the model with .fit() function. Test data is predicted with .predict() function and saved in new variable. For evaluating the model, confusion matrix and classification report is done.

DESITION TREE

```
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

Activity 2: Random forest model

A function named randomForest is created and train and test data are passed as the parameters. Inside the function, RandomForestClassifier algorithm is initialized and training data is passed to the model with .fit() function. Test data is predicted with .predict() function and saved in new variable. For evaluating the model, confusion matrix and classification report is done.

```
RANDOM FOREST

rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

RandomForestClassifier
RandomForestClassifier(random_state=42)
```

Activity 3: KNN

A function named ExtraTree is created and train and test data are passed as the parameters. Inside the function, ExtraTreeClassifier algorithm is initialised and training data is passed to the model with the .fit() function. Test data is predicted with .predict() function and saved in a new variable. For evaluating the model, a confusion matrix and classification report is done.

```
KNN

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

KNeighborsClassifier
KNeighborsClassifier()
```

Activity 5: Xgboost model

A function named xgboost is created and train and test data are passed as the parameters. Inside the function, GradientBoostingClassifier algorithm is initialized and training data is passed to the model with .fit() function. Test data is predicted with .predict() function and saved in new variable. For evaluating the model, confusion matrix and classification report is done.

```
XG BOOST

xg_clas = XGBClassifier(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=4,
    random_state=42
)
xg_clas.fit(X_train, y_train)

XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.1, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=4,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=100,
              n_jobs=None, num_parallel_tree=None, ...)
```

Now let’s see the performance of all the models and save the best model

Activity 5: Compare the model

```
Comparing the models
[ ]
print("KNN:", accuracy_score(y_test,p1 ))
print("Desition Tree:", accuracy_score(y_test,p2 ))
print("Random Forest:", accuracy_score(y_test,p3 ))
print("Xg Boost:", accuracy_score(y_test,p4 ))

KNN: 0.896551724137931
Desition Tree: 0.9655172413793104
Random Forest: 0.9655172413793104
Xg Boost: 0.9655172413793104
```

the results of models are displayed as output. From the four model Xgboost is performing well. From the below image, We can see the accuracy of the model. Xgboost is giving the accuracy of 96.5%.so we considering xgboost and deploying this model.

Activity 6: Evaluating performance of the model and saving the model

```
from keras.callbacks import EarlyStopping
from keras.optimizers import Adam
opt = Adam(lr=0.0001)

# Assuming you have defined your VGG16 model as vgg16

# Define Early Stopping callback
early_stopping = EarlyStopping(monitor='val_accuracy', patience=3, restore_best_weights=True)

# Compile the model (you may have already done this)
vgg16.compile(optimizer = "Adam" , loss='categorical_crossentropy', metrics=['accuracy'])


# # Train the model with early stopping callback
history = vgg16.fit(train, validation_data=test,
                    epochs=15,
                   steps_per_epoch=20,
                  callbacks=[early_stopping])
```

```
WARNING:absl:`lr` is deprecated in Keras optimizer, please use `learning_rate` or use the legacy optimizer, e.g.,tf.keras.optimizers.legacy.Adam.
Epoch 1/15
20/20 [==============================] - 28s 1s/step - loss: 1.4851 - accuracy: 0.6100 - val_loss: 1.3616 - val_accuracy: 0.6390
Epoch 2/15
20/20 [==============================] - 25s 1s/step - loss: 1.2048 - accuracy: 0.6750 - val_loss: 1.4731 - val_accuracy: 0.6336
Epoch 3/15
20/20 [==============================] - 31s 2s/step - loss: 1.1071 - accuracy: 0.7075 - val_loss: 1.6028 - val_accuracy: 0.6023
Epoch 4/15
20/20 [==============================] - 25s 1s/step - loss: 0.8216 - accuracy: 0.7675 - val_loss: 1.1175 - val_accuracy: 0.6979
Epoch 5/15
20/20 [==============================] - 26s 1s/step - loss: 0.7029 - accuracy: 0.7875 - val_loss: 1.2511 - val_accuracy: 0.6836
Epoch 6/15
20/20 [==============================] - 26s 1s/step - loss: 0.8559 - accuracy: 0.7575 - val_loss: 1.2702 - val_accuracy: 0.6685
Epoch 7/15
20/20 [==============================] - 26s 1s/step - loss: 0.6845 - accuracy: 0.8100 - val_loss: 1.0867 - val_accuracy: 0.7265
Epoch 8/15
20/20 [==============================] - 25s 1s/step - loss: 0.5509 - accuracy: 0.8325 - val_loss: 1.2089 - val_accuracy: 0.7069
Epoch 9/15
20/20 [==============================] - 27s 1s/step - loss: 0.5796 - accuracy: 0.8400 - val_loss: 0.8013 - val_accuracy: 0.7802
Epoch 10/15
20/20 [==============================] - 25s 1s/step - loss: 0.4136 - accuracy: 0.8850 - val_loss: 0.9262 - val_accuracy: 0.7560
Epoch 11/15
20/20 [==============================] - 26s 1s/step - loss: 0.4959 - accuracy: 0.8575 - val_loss: 1.0332 - val_accuracy: 0.7292
Epoch 12/15
20/20 [==============================] - 27s 1s/step - loss: 0.5788 - accuracy: 0.8300 - val_loss: 0.8914 - val_accuracy: 0.7587
```

```
vgg16.save('healthy_vs_rotten')
```

Note: To understand cross validation, refer this link. https://towardsdatascience.com/cross-validation-explained-evaluating-estimator-performance-e51e5430ff85.

BEST-FIT XgBoost

SAVING XGBoost

```
[ ]
import joblib
joblib.dump(xg_clas, "xg_flood_model.pkl")

['xg_flood_model.pkl']
```

```
[ ]
joblib.dump(features, "model_columns.pkl")

['model_columns.pkl']
```

Save Scaler Separately

```
[ ]
joblib.dump(scaler, "xg_scaler.pkl")

['xg_scaler.pkl']
```

Milestone 5: Application Building

In this section, we will be building a web application that is integrated to the model we built. A UI is provided for the uses where he has to enter the values for predictions. The enter values are given to the saved model and prediction is showcased on the UI.

This section has the following tasks

- Building HTML Pages
- Building serverside script

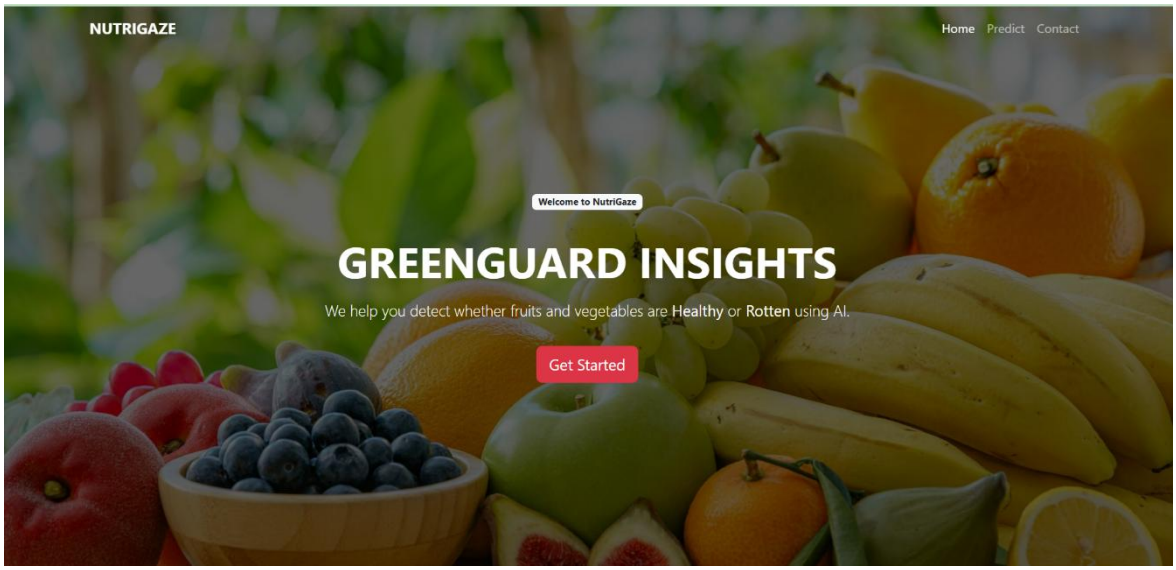
Activity1: Building Html Pages:

For this project create three HTML files namely

- index.html
- predict.html
- base.html

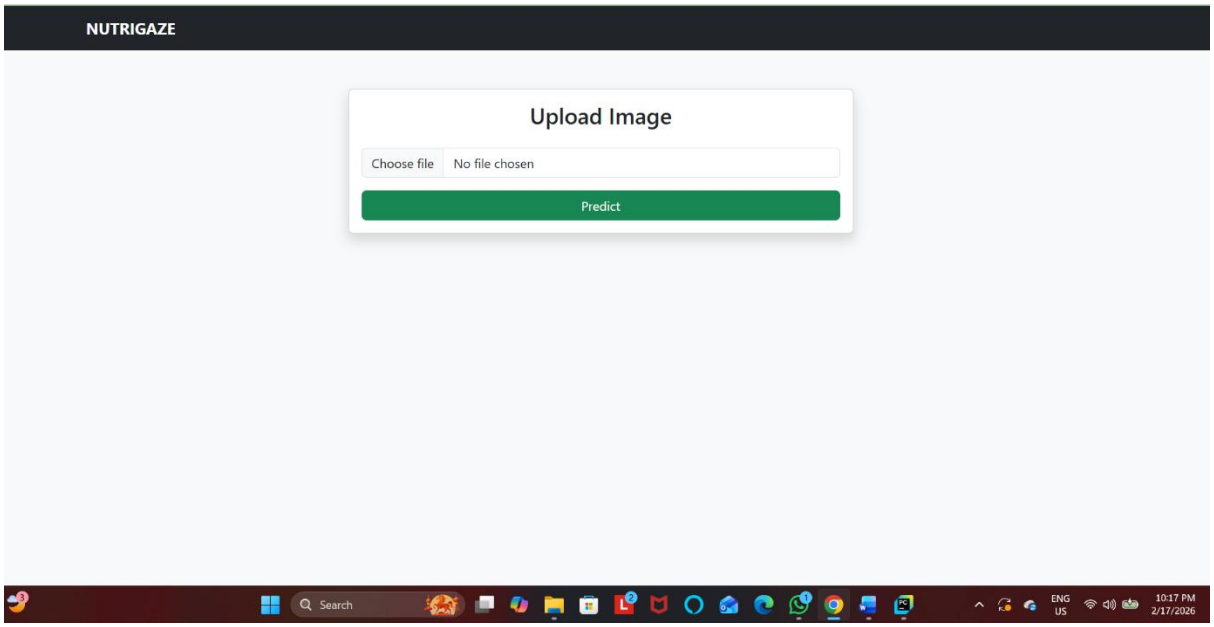
and save them in templates folder.

Let's see how our home.html page looks like:



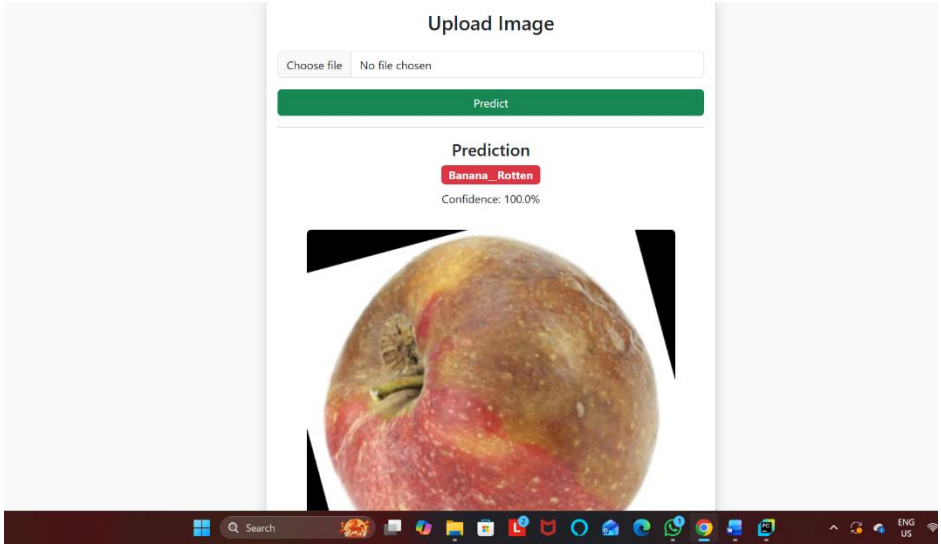
Now when you click on \Predict Flood button from top right corner you will get redirected to predict.html

Lets look how our predict.html file looks like:



Now when you click on detect button a popup will shown weather the Flood is Expected or Not based on the entered Data.

Lets look how our Pop up looks like:



Activity 2: Build Python code:

Import the libraries

```
from flask import Flask, render_template, request
import numpy as np
import joblib
```

Load the saved model. Importing flask module in the project is mandatory. An object of Flask class is our WSGI application. Flask constructor takes the name of the current module (__name__) as argument.

Here we will be using declared constructor to route to the HTML page which we have created earlier.

In the example, '/' URL is bound with index.html function. Hence, when the home page of the web server is opened in browser, the html page will be rendered. Whenever you enter the values from the html page the values can be retrieved using POST Method.

Retrieves the value from UI:

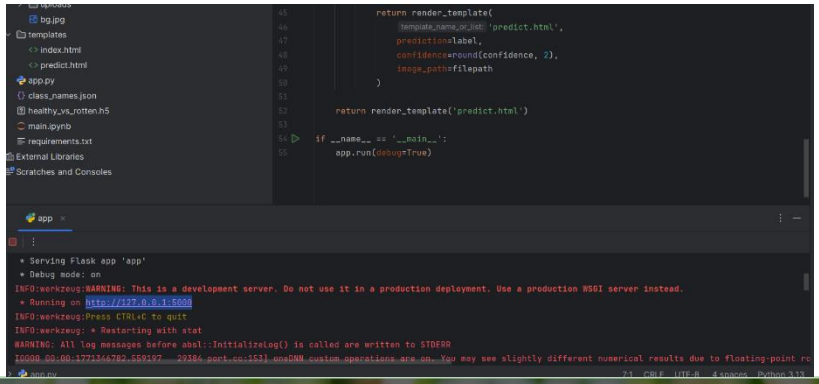
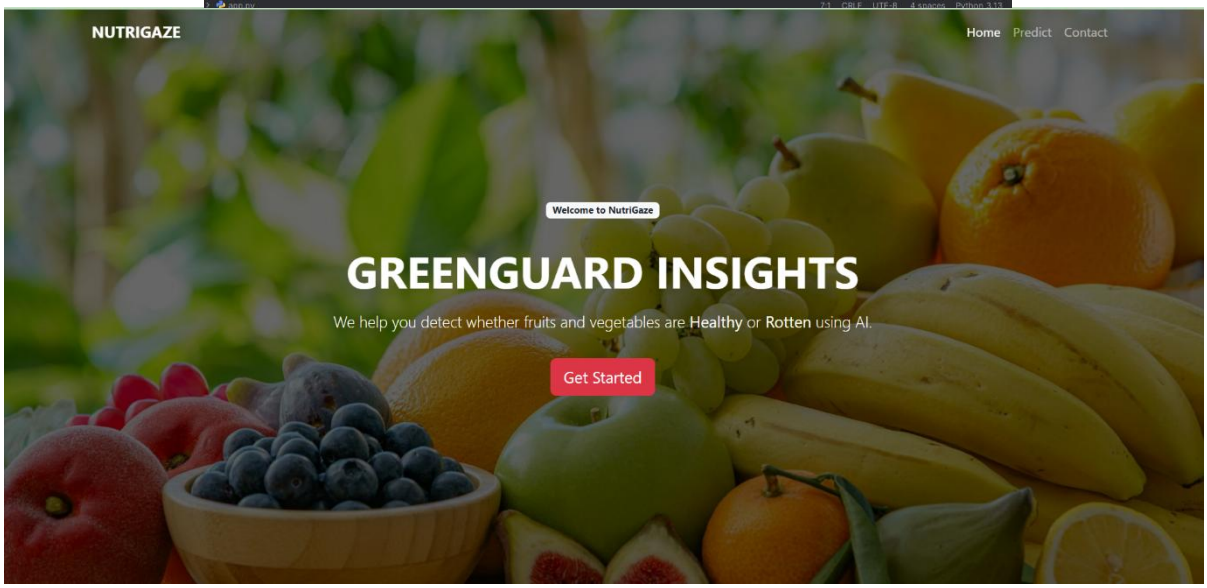
```python
app = Flask(__name__)

model = joblib.load("xg_flood_model.pkl")
xg_scaler = joblib.load("xg_scaler.pkl")

@app.route('/')
def home():
    return render_template('home.html')

@app.route('/predict_page')
def predict_page():
    return render_template('predict.html')

@app.route('/predict', methods=['POST'])
def predict():

    data = [
        float(request.form['cloud']),
        float(request.form['annual']),
        float(request.form['janfeb']),
        float(request.form['marchmay']),
        float(request.form['junsep'])
    ]
```

Here we are routing our app to predict() function. This function retrieves all the values from the HTML page using Post request. That is stored in an array. This array is passed to the model.predict() function. This function returns the prediction. And this prediction value will rendered to the text that we have mentioned in the submit.html page earlier.

Main Function:

```python
    data = [
        float(request.form['cloud']),
        float(request.form['annual']),
        float(request.form['janfeb']),
        float(request.form['marchmay']),
        float(request.form['junsep'])
    ]

    df = np.array([data])
    scaled = xg_scaler.transform(df)
    pred = model.predict(scaled)[0]

    result = "Flood Expected" if pred == 1 else "No Flood"

    return render_template('predict.html', prediction=result)

if __name__ == "__main__":
    app.run(debug=True)
```

Activity 3: Run the application

- Open anaconda prompt from the start menu
- Navigate to the folder where your python script is.
- Now type “python app.py” command
- Navigate to the localhost where you can view your web page.

- Click on the predict button from top left corner, enter the inputs, click on the submit button, and see the result/prediction on the web.